Fiscal Year:	FY 2024	**Task Last Updated:**	FY 11/06/2025
PI Name:	Nemec-Bakk, Ashley Ph.D.		
Project Title:	The Use of Two New Ground-Based Models of Deep Space Travel to Study the Role of Mitochondria and Oxidative Stress in Cardiovascular Effects		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 TRISH-RFA-2201-PD: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Boerma, Marjan Ph.D. (MENTOR: University of Arkansas, Little Rock)		
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Task Description:	POSTDOCTORAL FELLOWSHIP On future missions into deep space, astronauts will be subject to chronic low doses of ionizing radiation. Since only a small number of astronauts have traveled outside of low-Earth orbit, it is difficult to estimate the physiological risk of long distance space travel on individual organ systems. Therefore, studies determining the effects of space radiation rely on animal models to estimate the risks of degenerative health effects and to understand biological mechanisms involved. This project addresses three knowledge gaps in this area of research: 1) There is a lack of studies that investigate the combination of galactic cosmic radiation (GCR) and microgravity on the cardiovascular system and risks for degenerative diseases; 2) Biological and pathological pathways of degenerative cardiovascular diseases due to space radiation and microgravity are not yet fully understood; and 3) Although there has been an increase in woman astronauts since 1950, most ground-based studies have been performed on males. To address these knowledge gaps, this project will make use of two ground-based models in which male and female C57BL/6J mice will be subjected to the modeled deep space environment with two methods that are being developed in our laboratory. The first model makes use of the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory to create a mixed ion field simulating GCR. The second model makes use of a chronic gamma ray facility at our home institution to expose mice to chronic low-dose rate ionizing radiation for one month. In both models, we will combine radiation exposure with simultaneous hind limb unloading (HLU) to model fluid shifts in microgravity, thereby simulating the environment in deep space. The project tests the hypothesis that exposure to simulated GCR or chronic low-dose rate exposures to ionizing radiation will result in mitochondrial damage and increased oxidative stress, which will have an adverse impact on the function of the cardiovascular system. This hypothesis will be tested in the following aims: Aim 1: Determine the effects of simulated GCR and HLU on mitochondria, oxidative stress, and function of the heart, major arteries, and microvasculature in a mouse model. The influence of sex on cardiovascular outcomes of simulated GCR and HLU will be assessed. Aim 2: Test the effectiveness of secoisolariciresinol diglucoside, a natural antioxidant, as a potential countermeasure against cardiovascular effects of chronic low-dose gamma ray exposure and HLU. Cardiac function will be measured with high-resolution echocardiography, and the eye structure and retinal vasculature will be imaged with optical coherence tomography. Arterial function will be measured with ex vivo perfused aorta obtained from the mice at the end of the studies. Lastly, tissue samples will be collected to measure mitochondrial function and antioxidant enzyme activities. Taken together, these hypothesis-driven aims will fill crucial knowledge gaps that are relevant to NASA's future long duration space flights beyond low-Earth orbit. This study will elucidate if and how GCR and microgravity can initiate cardiovascular degenerative diseases and whether there is a different risk between males and females.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	On future missions into deep space, astronauts will be subject to chronic low doses of ionizing radiation. Since only a small number of astronauts have traveled outside of low Earth orbit, it is difficult to estimate the physiological risk of long-distance space travel on individual organ systems. Therefore, studies determining the effects of space radiation rely on animal models to estimate the risks of degenerative health effects and to understand biological mechanisms involved. This project addresses three knowledge gaps in this area of research: 1) There is a lack of studies that investigate the combination of galactic cosmic radiation (GCR) and microgravity on the cardiovascular system and risks for degenerative diseases; 2) Biological and pathological pathways of degenerative cardiovascular diseases due to space radiation and microgravity are not yet fully understood; and 3) Although there has been an increase in woman astronauts since 1950, most ground based studies have been performed in males. To address these knowledge gaps, this project will make use of two ground-based models in which male and female C57BL/6J mice will be subjected to the modeled deep space environment with two methods that are being developed in our laboratory. The first model makes use of the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory to create a mixed ion field simulating GCR. The second model makes use of a chronic gamma ray facility at our home institution to expose mice to chronic low dose rate ionizing radiation for one month. In both models, we will combine radiation exposure with simultaneous hind limb unloading (HLU) to model fluid shifts in microgravity, thereby simulating the environment in deep space. The project tests the hypothesis that exposure to simulated GCR or chronic low-dose rate exposures to ionizing radiation will result in mitochondrial damage and increased oxidative stress, which will have an adverse impact on the function of the cardiovascular system. This hypothesis will be tested in the following aims: Aim 1: Determine the effects of simulated GCR and HLU on mitochondria, oxidative stress and function of the heart, major arteries and microvasculature in a mouse model. The influence of sex on cardiovascular outcomes of simulated GCR and HLU will be assessed. Aim 2: Test the effectiveness of secoisolariciresinol diglucoside (SDG), a natural antioxidant, as a potential countermeasure against cardiovascular effects of chronic low-dose gamma ray exposure and HLU. Cardiac function will be measured with high-resolution echocardiography, and the eye structure and retinal vasculature will be imaged with optical coherence tomography. Lastly, tissue samples will be collected to measure mitochondrial function and antioxidant enzyme activities. Taken together, these hypothesis-driven aims will fill crucial knowledge gaps that are relevant to NASA's future long-duration space flights beyond low Earth orbit. This study will elucidate if and how GCR and microgravity can initiate cardiovascular degenerative diseases and whether there is a different risk between males and females. GCR+HLU were found to have a compounding effect on the cardiovascular system and retina. The cardiovascular system was affected differently in male and female mice. Additionally, only the female retina demonstrated structural changes which could indicate the start of a disease state. These adverse changes were only seen in the groups receiving the higher dose of radiation (1.5 Gy GCR) with simultaneous HLU. The groups receiving a mission relevant dose (0.5 Gy GCR) did not show indications of adverse effects. Due to space limitations and time constraints, only male mice were used in aim 2. 2 groups of 40 male C57BL/6J mice

	were housed around an open field cesium source for 30 days. Subcohorts of these animals underwent HLU to determine if HLU and radiation exposure had a compounding effect on the cardiovascular system. 1 group of animals were sacrificed immediately after the 30-day experiment to determine acute effects of the space environment, and the second group was followed for up to 6 months to determine the chronic cardiovascular effects. HLU groups demonstrated decreased body weight after the 30 days of HLU but returned to control values after returning to weight bearing. Echocardiography was performed at 6 months after radiation and HLU, which showed the combination group had significantly reduced cardiac output and ejection fraction. Histological analysis of cardiac tissues demonstrated no change in collagen content at the acute timepoint, but there was a significant increase in collagens in the combination group 6 months after exposure. All treatment groups expressed an increase in intracellular 4-Hydroxynonenal, a stable byproduct of lipid peroxidation, compared to control indicating increased oxidative stress. Production of a synthetic SDG product was completed at the end of the project and will be tested in future experiments. The above findings suggest the combination of radiation and HLU result in degenerative effects in the cardiovascular system and the retina, but these adverse effects are dependent on the radiation dose and time of exposure. Moreover, males and females are affected through different mechanisms and these differences should be studied further.
Bibliography Type:	Description: (Last Updated: 11/06/2025)
Articles in Peer-reviewed Journals	Nemec-Bakk AS, Sridharan V, Willey JS, Koturbash I, Williams DK, Chesal M, Patel CM, Borg AM, Reno K, Gifford G, Newhauser W, Williams J, Chancellor JC, Boerma M. "Sex-specific effects on the heart from combined exposure to simulated galactic cosmic radiation and hindlimb unloading." Life Sci Space Res. 2025 Feb:44:38-46. https://doi.org/10.1016/j.lssr.2024.12.001 ; PMID: 39864910 , Feb-2025